

PROBLEMS CLASS 2: SOLUTIONS

1. Phase portrait recipe

Linear stability analysis is an algebraic method to determine the stability of a system's equilibria. Phase portraits are a graphical method which can often also imply the same stability. Think of them as complementary tools. A recipe for drawing a phase portrait is:

- (i) Compute and draw equilibria, indicating stability from a linear stability analysis if you have done one.
- (ii) Draw nullclines ($\dot{x} = 0$, $\dot{y} = 0$):
 - Trajectories along $\dot{x} = 0$ will be vertical,
 - Trajectories along $\dot{y} = 0$ will be horizontal.

Equilibria will be where $\dot{x} = 0$ and $\dot{y} = 0$ nullclines intersect.

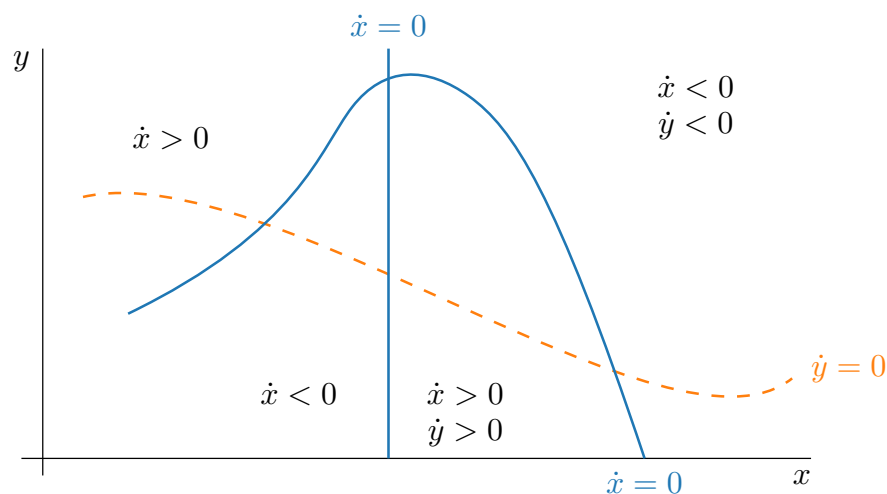
- (iii) Test the regions between the nullclines for $\dot{x}, \dot{y} \gtrless 0$.
- (iv) Infer direction of trajectories and join up.

Trajectories in a phase portrait do not intersect: can you explain why?

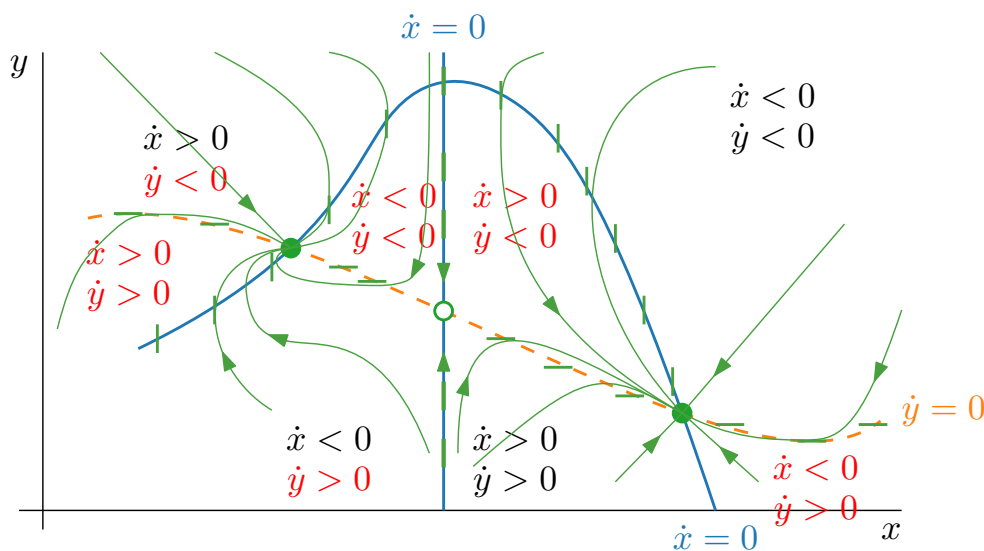
Solution: Trajectories in a phase portrait do not intersect, because where they intersect, the trajectory (i.e. value of $(\dot{x}, \dot{y})$) should be unique in a deterministic (i.e. non-random) system.

2. Trajectory drawing practice

Consider the half-completed phase portrait below for the system (x, y) . Identify the equilibria and complete the testing of the regions. Then draw in the trajectories and infer the stability of the equilibria.



Solution:



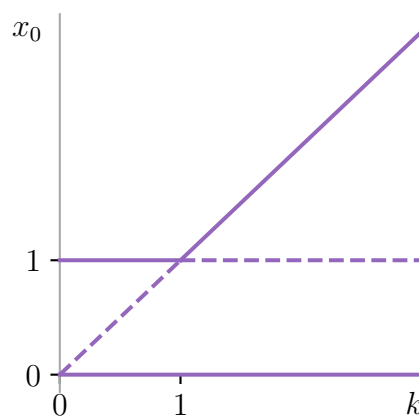
3. Bifurcation investigation (Exam 2023)

Consider the bifurcation diagram on the right, which shows how the equilibria x_0 of a single-species population model depend on a single parameter, k :

- Write down all the equilibria, x_0 , shown on the graph.
- Hence write down a model for a population $x(t)$ in the form

$$\frac{dx}{dt} = f(x, k)$$

which would produce this bifurcation diagram.



Solution: For each value of k , there are three values of x_0 :

$$x_0 = 0, \quad k, \quad \text{and } 1.$$

This suggests an equation of the form

$$\frac{dx}{dt} = -x(x - k)(x - 1),$$

i.e. the Allee model seen in class. Drawing the portrait confirms the stability.

4. Competitive Lotka–Volterra

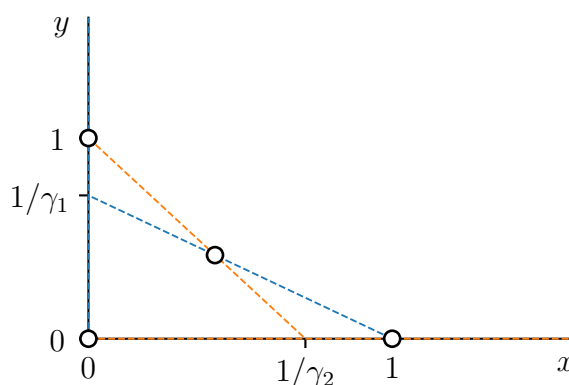
Recall from last week's lectures that the competitive Lotka–Volterra system, in nondimensional form, looks like

$$\frac{dx}{dt} = x(1 - x - \gamma_1 y), \quad (1)$$

$$\frac{dy}{dt} = \beta y(1 - y - \gamma_2 x), \quad (2)$$

with $\beta, \gamma_1, \gamma_2 > 0$.

Drawing on the nullclines and equilibria for the $\gamma_1 > 1, \gamma_2 > 1$ gives us

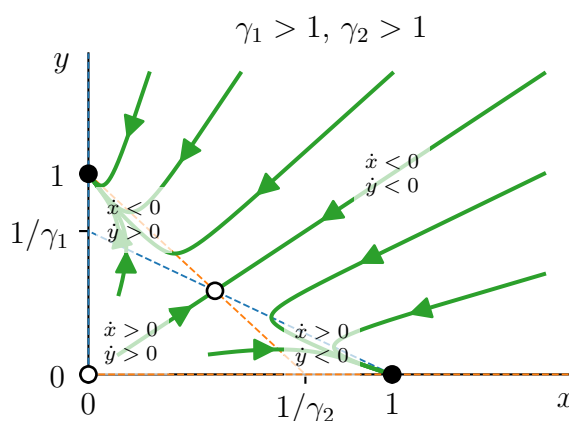


- Why is there no equilibrium where nullclines intersect at $(0, 1/\gamma_1)$?
- Test the regions between the nullclines for $\dot{x}, \dot{y} \geq 0$.
- Complete the phase portrait: infer the trajectories and further infer the stability of the equilibria, which you should colour in accordingly.

You can compare your trajectories to those in the completed diagram in the course notes or on last week's handout.

Solution:

- Equilibria only exist where $\dot{x} = 0$ and $\dot{y} = 0$ lines intersect.
- Completed diagram looks like:



5. Problem sheet 1: variations

Consider this altered system from problem sheet 1:

$$\frac{dx}{dt} = x - 2y + 10, \quad (3)$$

$$\frac{dy}{dt} = 2x + 2y + 4. \quad (4)$$

Draw a phase portrait for this system and, from the phase portrait alone, infer the stability of any equilibria.

Solution: The nullclines for $\dot{x}$ and $\dot{y}$ are

$$y = \frac{x+10}{2} \quad \text{and} \quad y = -x-2, \quad (5)$$

and the phase plane looks like

